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Independent Components in Dynamic Refraction

Marina Alterman, Yoav Y. Schechner, Pietro Perona and Joseph Shamir

Abstract

Refraction causes random dynamic distortions in atmospheric turbulence and in views across a water interface. The later scenario is experienced by submerged animals seeking to detect prey or avoid predators, which may be airborne or on land. Man encounters this when surveying a scene by a submarine or divers, while wishing to avoid the use of an attention-drawing periscope. The problem of inverting random refracted dynamic distortions is difficult, particularly when some of the objects in the field of view are moving. On the other hand, in many cases, just those moving objects are of interest, as they reveal animal, human or machine activity. Furthermore, detecting and tracking these objects does not necessitate handling the difficult task of complete recovery of the scene. We show that moving objects can be detected very simply, with low false-positive rates, even when the distortions are very strong and dominate the object motion. Moreover, the moving object can be detected even if it has zero mean motion. While the object and distortion motions are random and unknown, they are mutually independent. This is expressed by a simple motion feature which enables discrimination of moving object points vs. the background.

Index Terms

- *Marina Alterman, Yoav Schechner and Joseph Shamir are with the Dept. of Elect. Eng., Technion - Israel Inst. of Technology, Haifa 32000, Israel.*
- *Pietro Perona is with the Dept. of Electrical Engineering, California Institute of Technology, Pasadena, CA 91125.*
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